

Academic Statement of Purpose

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Growing up, I often witnessed my grandparents struggling with technology. Sometimes, it was something as simple as turning on the television; other times, it was much more critical. On several occasions while driving, not knowing how to use a GPS meant a few wrong turns left them lost for hours. Thankfully, nothing serious ever happened, but each situation was frightening, especially because it was preventable. The tools to keep them safe were there, but they were inaccessible to people who struggled to interact with them. These experiences made me realize that technology is only as powerful as its accessibility.

This perspective is what draws me towards fields like artificial intelligence (AI) and extended reality (XR). Both of these technologies hold the promise of bridging the gap between powerful computational systems and human accessibility. AI tools, such as natural language chatbots, simulate the experience of talking to another person, while XR applications recreate real-life environments in immersive ways that feel natural and intuitive. Together, these fields have the potential to make technology more approachable, effective, and impactful for people of all backgrounds. Pursuing graduate study through SUGS will give me the technical depth in AI and XR to launch a career in software engineering, where I can design innovative, human-centered tools that improve people's daily lives.

My academic foundation has been shaped by a Computer Science major with minors in Business and User Experience (UX) Design. This combination has given me a well-rounded perspective: technical rigor from EECS courses, organizational and strategic thinking from business, and a human-centered design mindset from UX. These experiences have taught me that developing technology is not just about algorithms and systems; it is also about anticipating how people will interact with these tools and ensuring accessibility for all users.

As an undergraduate, I have consistently sought out challenging courses that require both depth of understanding and resilience, skills needed for graduate coursework. In EECS 482 (Operating Systems), I learned the importance of carefully dissecting specifications and thoroughly planning before implementing code, which sharpened my problem-solving and system design skills. I am currently taking EECS 442 (Computer Vision), which is strengthening my foundation in AI and serving as a bridge to exploring how computer vision and XR intersect. Beyond coursework, I have also been contributing as an Instructional Aide for EECS 481 (Software Engineering) since the Winter 2025 semester. In this role, I host office hours, grade exams, develop grading automation, and support students in understanding industry-level concepts and research. Given the opportunity, I plan to continue contributing as a Graduate Student Instructor for EECS 481 under Xinyu Wang and Wes Weimer during my SUGS tenure. This would not only reinforce my

own knowledge of software engineering but also allow me to mentor the next generation of students, an experience I find incredibly rewarding.

My academic pursuits have been enriched by hands-on research and project experiences. My most formative experience has been with CLAWS, an interdisciplinary project team competing in NASA's SUITS Challenge. Since my freshman year, I have worked on developing augmented reality (AR) interfaces for NASA astronauts on lunar missions and presented our projects at the NASA Johnson Space Center in Houston. Using Unity, MRTK, and the HoloLens 2, I developed features that guided astronauts through navigation and geosampling tasks in lunar environments. Over the years, I transitioned from AR feature development to serving as AI lead and executive board member, coordinating integration across development teams. This work required not only technical skill but also research into astronaut workflows and cross-team collaboration. While I am already taking steps to learn more about AI by joining the AI subteam, SUGS will provide me with greater educational opportunities to contribute to the team while also enabling me to continue this work and mentor younger members.

In addition, I gained formal research experience as an AR Research Assistant in the SURE program for the MAVRIC team during Summer 2023. To help the team research shared mental models in human-robot collaboration, I developed an AR color-sorting task on the Magic Leap 2, utilizing hand and eye tracking APIs and ROS2 to coordinate teamwork between human and robot agents. This experience not only improved my technical expertise in AR platforms but also taught me how to frame engineering work in the context of human-centered research questions. Writing research reports and documenting results gave me valuable practice in academic communication and further convinced me of the importance of bridging technical systems with human interaction.

Looking ahead, I aspire to become a software engineer who creates innovative systems that harness the potential of AI and XR to improve people's daily lives. I am particularly excited about the intersection of AI and XR, and how these technologies can be integrated to redefine human-computer interaction. I am eager to learn from faculty such as Professor Jiasi Chen, whose ECE 598 course on AI-Enabled Mixed Reality directly aligns with my aspirations. At the same time, I want to broaden my knowledge in other areas of computer science, such as distributed systems and databases, which I was unable to fit into my undergraduate schedule. By pursuing SUGS, I will also gain an additional summer to pursue internship opportunities, preparing me to enter the workforce with greater academic and industry experience.

My personal experiences, academic foundation, and research engagements have motivated me to pursue graduate study in CSE. The SUGS program is the ideal pathway for me to continue exploring innovative technologies while broadening my knowledge across computer science. With this program, I will be able to deepen my technical expertise, stay engaged with Michigan's

community, and pursue a software engineering career creating human-centered technologies that make lasting differences in people's lives.